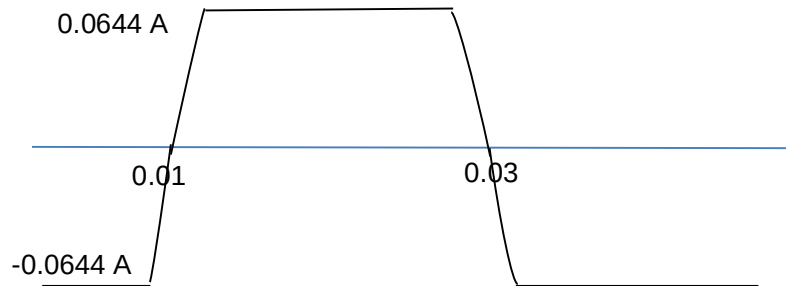


- 5 (a) The induced e.m.f. is given by the rate of change of flux linkage.

(b) (i) Induced e.m.f.  $= \frac{d\Phi}{dt} = NA \frac{dB}{dt} = 500 \times 7.25 \times 10^{-4} \frac{2.4 \times 10^{-2}}{0.009} = 0.967 \text{ V}$

Maximum current  $= \frac{0.967}{10.0 + 5.00} = 6.44 \times 10^{-2} \text{ A}$

(ii)



- 6 (a) (i)  $A=60$